Additive Manufacturing of Magnetic Components for Power Electronics Integration

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***Abstract*—In an effort to simplify power electronics integration, we explored an additive manufacturing (AM) process, or commonly known as three-dimensional (3D) printing for fabricating magnetic components. A commercial fused-deposition-model (FDM) 3D printer was modified to extrude pastes of powder materials into 3D structures. We formulated a low-temperature curable Poly-Mag paste, which is a mixture of magnetic powder and a polymer, as a feed material for printing the magnetic core and used a commercial nanosilver paste for printing the winding. After the structure was printed, it was cured at 250ºC for an hour without any external pressure to form the magnetic component. The electrical resistivity of the printed winding was about three times higher than that of bulk silver. This is because of the low curing temperature resulted in a lower density than the bulk. The relative permeability dispersion of the printed core had a bandwidth up to 10 MHz with a static relative permeability of about 10. Both the winding and core magnetic properties will be improved by adjusting the feed paste formulations and their flow characteristics and fine-tuning the printer parameters, such as motor speeds, extrusion rate, and nozzle sizes.**

Keywords—additive manufacturing; power electronics integration; magnetic components; magnetic pastes.

I. INTRODUCTION

Integration of power inductors and transformers as shown in Figure 1(a) is challenging due to large core volume or unconventional geometries to distribute flux or control coupling [1-5]. Commercial magnetic components consist of cores and windings are fabricated separately and assembled as shown in Figure 1(b). This manufacturing approach gives rise to bulky discrete components and precludes the implementation of high-density, high-performance integrated geometries. For example, the uniform-flux magnetic component structures [6] shown in Figure 1(c) cannot be fabricated by the current manufacturing process. Thus, to improve power conversion efficiency, power density, and reliability of a power electronics system, there is a need for manufacturing technologies that would enable ease of integration of the magnetic components.

Additive manufacturing (AM) or three-dimensional (3D) printing is a layer-by-layer process of making products and components from a digital model. Its potential has been demonstrated for applications in various industries. Some key benefits of AM are shorter lead times, mass customization, reduced parts count, more complex shapes, less material waste, and lower life-cycle energy use [7]. Recently, some research groups explored application of AM in power electronics. For example, Oak Ridge National Laboratory (ORNL) 3D-printed an aluminum heat sink and a plastic lead frame for a 10-kW power inverter [8]; Wei Liang et.al. 3D-printed a plastic structure as a mold for making a unique sterling-silver inductor [9]; and Proto-pasta introduced an iron-powder filled PLA filament for use with a fused-deposition-model (FDM) 3D printer to make magnetic cores with relative permeability of about 1.5 [10]. Thus far, we have not found any published work in the literature that 3D-printed both magnetic and metal materials to form magnetic components.

The purpose of this work is to explore the feasibility of using a 3D-printing process for fabricating magnetic components. A commercial FDM 3D printer was modified to extrude pastes of powder materials into 3D structures. A low-temperature curable Poly-Mag paste was formulated to serve as a feed material for the magnetic core, while our patented nanosilver paste was used for the winding. After the magnetic structures were printed, a low-temperature process was used to cure the Poly-Mag core and partially sinter the winding. The resistivity of the winding and the complex permeability of the magnetic core were characterized.

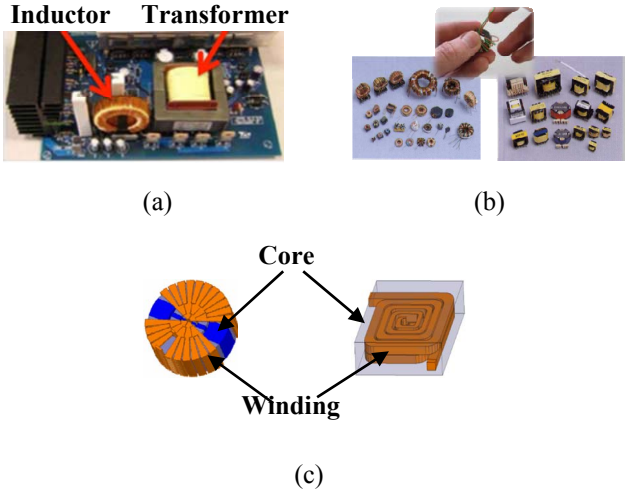


Figure 1 (a) Typical power supply consisting of a number of discrete magnetic components; (b) basic process for making the windings of inductors and transformers; and (c) integrated uniform-flux inductor design.

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II. Experimental Precedure

A. *Custom-modified paste extrusion-based 3D printer*

We chose a paste extrusion-based 3D printer in this study because it offers ease and flexibility of co-processing multiple materials [11]. Use of paste as the feed material also reduces material waste, lowers the equipment cost, and simplifies the part-construction process. The paste-based additive process can be readily scaled-up to manufacture a multi-material, multi-functional system by inserting multiple paste materials into a feed-cartridge assembly. Thus, this process platform offers the potential for further integration of a power electronics circuit by concurrent manufacturing of capacitive, magnetic, and resistive components. We purchased a low-cost FDM 3D printer from Ultimaker and modified it to co-dispense pastes stored in two separate syringes. The two-syringe paste-extrusion 3D printer is shown in Figure 2.

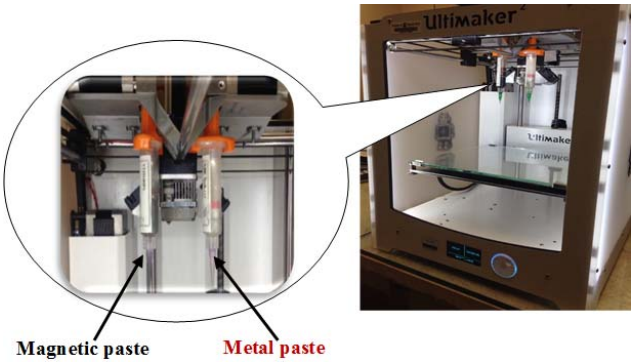


Figure 2 The modified FDM 3D printer for dispensing pastes stored in two separate syringes.

B. Nanosilve paste

To print electrical winding of magnetic components, we tested our nanosilver paste developed for chip bonding [12], [13]. The material can be sintered at temperatures below 250oC to result in excellent thermal and electrical properties. Figure 3 shows that the nanosilver paste can be adapted to the 3D printing platform and extruded out to form different winding structures.

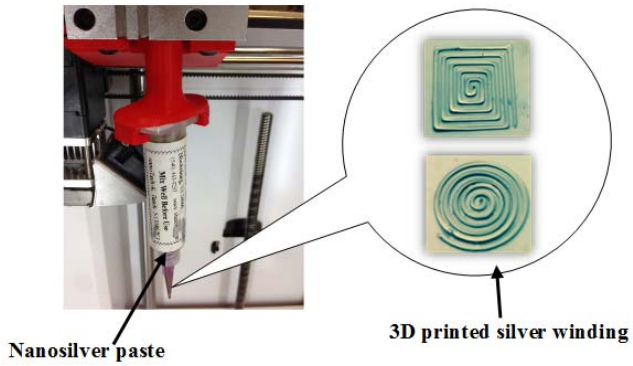


Figure 3 . Nanosilver paste for custom-modified paste-extrusion 3D printer

C. Poly-Mag paste

A low-temperature (<250oC) curable magnetic paste consisting of a soft magnetic powder and a thermoset polymer called Poly-Mag paste was formulated by extending our prior research experience in the development of the nanosilver paste [14]. A Permalloy powder (from ESPI Metals) with an average particle size of 12 μm were used in synthesizing the paste. Benzocyclobutene (BCB) which is a thermoset polymer obtained from Dow Chemical Company (Midland, MI) was added to the magnetic powder to serve as a binder. BCB is widely used in the electronics industry due to its ease of processing, low dielectric constant (2.7), and good thermal stability. Other organic components were added to produce the paste with good particle dispersion, flowability, and viscosity. Figure 4 illustrates some major ingredients in the Poly-Mag paste and a toroid core being-printed in the 3D-printer. To compare the quality of printed magnetics with that of fabricated by other processes, a toroid core of the same Poly-Mag composition was made by an injection-molding process.

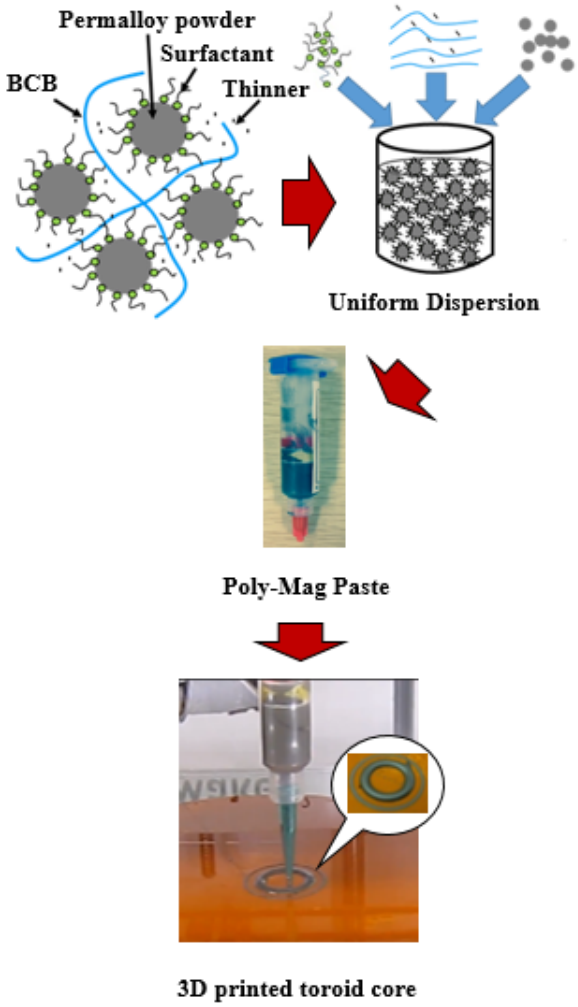


Figure 4 A low-temperature curable Poly-Mag paste for custom-modified paste-extrusion 3D printer

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D. 3D-printed magnetic components

Feasibility of the modified 3D printer for fabricating magnetic components was tested using the Poly-Mag paste as the feed material for the magnetic core and the nanosilver paste for the winding. A simple planar inductor was designed for quick evaluation of the co-extrusion capability of the printer. After 3D structures were fabricated in the printer, they were heated in a programmable muffle furnace to simultaneously cure the polymer in the magnetic core and sinter the nanosilver winding. Figure 5 shows the 3D-printing process to form a planar inductor, the heating profile used to harden the printed part, and a finished planar inductor.

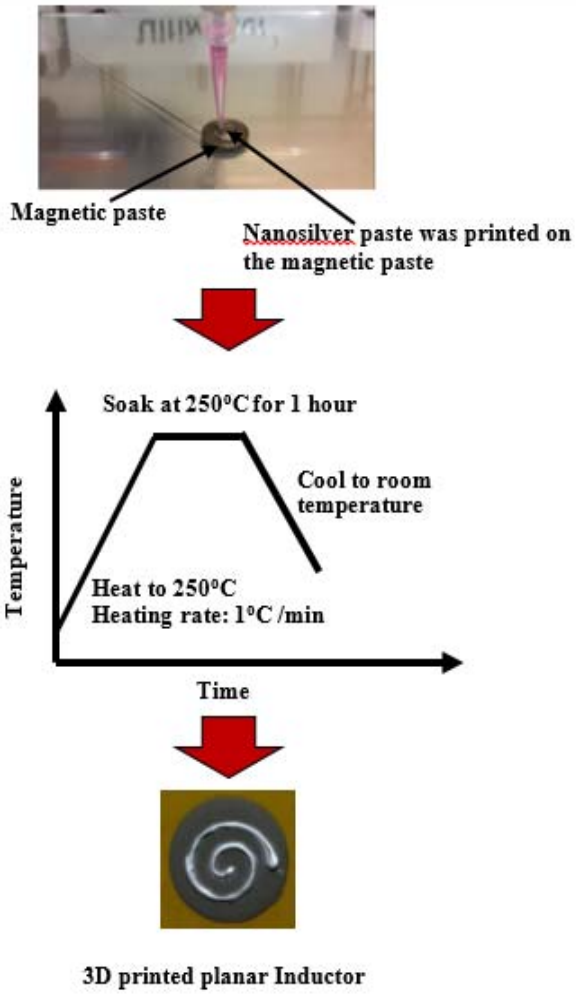


Figure 5 Process of 3D printing Planar Inductor

III. CHARACTERIZATION

A. Electrical resistivity of low-temperature sintered 3D-printed winding

Table 1 lists the electrical resistivity of a 3D-printed winding after low-temperature sintering the nanosilver paste. The electrical resistivity of the printed winding is about three times higher than that of the pure silver, but significantly lower than windings formed using other silver pastes under the same processing conditions, i.e. pressure-free, 250°C sintering. There are claims of achieving better electrical conductivity using inks of nanoparticles of silver or copper. However, these inks are only able to print conductive films with thickness up to one or two microns. In magnetic components for most of power electronics circuits, winding thickness needs to be at least tens of microns to lower the winding loss. The magnetic core thickness of the as-printed planar inductor is about 2mm and the winding layer thickness is about 0.8mm.

Table 1 Electrical resistivity of low temperature sintered 3D-printed winding and that of bulk silver

Material	Electrical Resistivity ($10^{-8}\Omega\cdot m$)
3D printed silver winding	4.8
Pure silver wire	1.6

B. Complex permeability dispersion spectra of the Poly-Mag core

Figure 6 is a plot of the complex permeability dispersion spectra (black lines) of a Poly-Mag toroid made by the 3D printer. Also plotted in the figure, in red, is the complex permeability dispersion spectra of an injection-molded Poly-Mag core. The relative permeability of the printed core is 9, only 10% lower than that of the injection-molded core. Both of the cores had stable real parts of permeability beyond 30MHz.

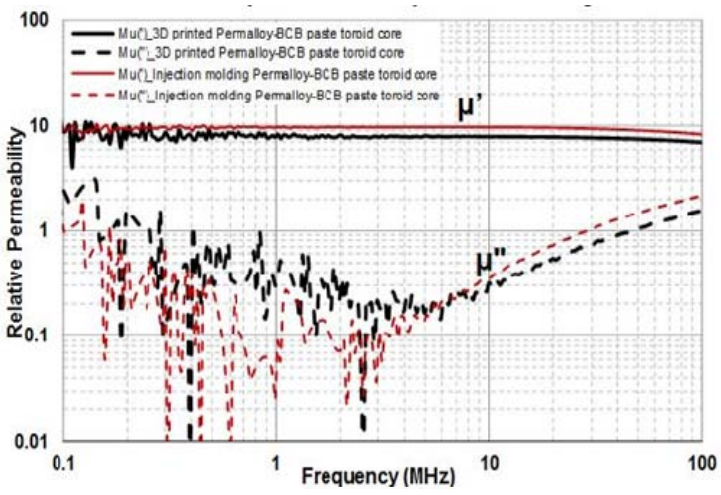


Figure 6 Comparison of complex permeability dispersion spectra of 3D printed Poly-Mag paste toroid core and injection molding Poly-Mag paste toroid core

C. Microstructure of 3D-printed Poly-Mag core

Figure 7 is a micrograph of a 3D-printed Poly-Mag core. The presence of some voids and cracks had the effect of distributed air gaps in the core, resulting in lower relative permeability as shown in Figure 6. The air trapped in the core may happen during the printing process. Therefore, the

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properties of the magnetic components are functions of both material chemistry and printing parameters.

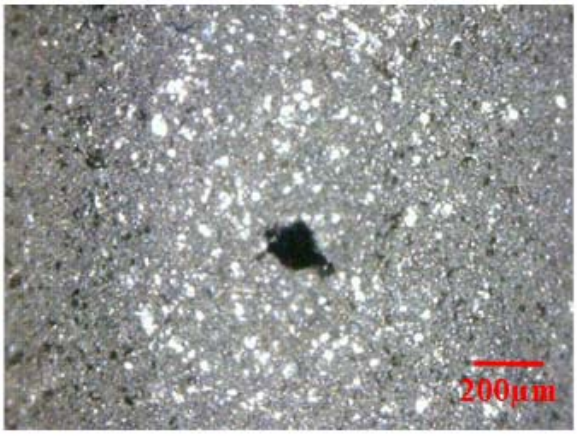


Figure 7 A micrograph of a 3D-printed Permalloy+Benzocyclobutene (BCB) core

IV. Summary and Future Work

We took a low-cost FDM (fused-deposition-model) 3D printer and successfully modified it to co-extrude two materials in the form of paste for additive manufacturing magnetics components. A low temperature curable Poly-Mag paste consisting of Permalloy powder and Benzocyclobutene (BCB), was formulated as the feed paste for the magnetic core, while a nanosilver paste was used for printing the winding. We demonstrated that the custom-modified 3D printer is capable of printing the two different pastes into 3D magnetic structures. Properties of the magnetic components are functions of material chemistry and printing parameters, which we will focus our future work to adjust in order to improve the component properties.

Acknowledgment

We acknowledge the financial support of a seed grant from Virginia Tech's Institute of Critical Technology and Applied Sciences. Y. Yan has been supported by High Density Integration Mini-consortium in the Center for Power Electronics at Virginia Tech. One of us, Y. Mei acknowledges the financial support from the Science and Technology Office of Tianjin Municipal Government.

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